Day 104: The Week 15 Build — A Small Retrieval-Augmented Question-Answering System

MD Mutasim Billah | 52-Week Data Science to ML/AI Roadmap | Week 15, Day 7 | Complete study guide

Watch: What Is RAG? Retrieval-Augmented Generation Explained Simply (Redis) | RAG Explained in 5 Minutes (KnowHive)

Script: `day104_rag_pipeline.py` · every number in this guide is copied by program from `day104_run_output.txt`, the real output of that script.

The week's six exercise parts, copied verbatim, assembled into retrieval-augmented question answering over every paragraph of SQuAD's validation set. Without passages, a small instruction-tuned T5 answers almost nothing (F1 0.043); with the top three retrieved passages it reaches 0.691, and an extractive reader 0.736. Handing over the right paragraph instead (the oracle) is worth another +0.103 to +0.131. When RAG fails completely, the cause splits roughly in half between retrieval and reading.

0. Where today fits

DEFINITION — start here, no prior knowledge assumed

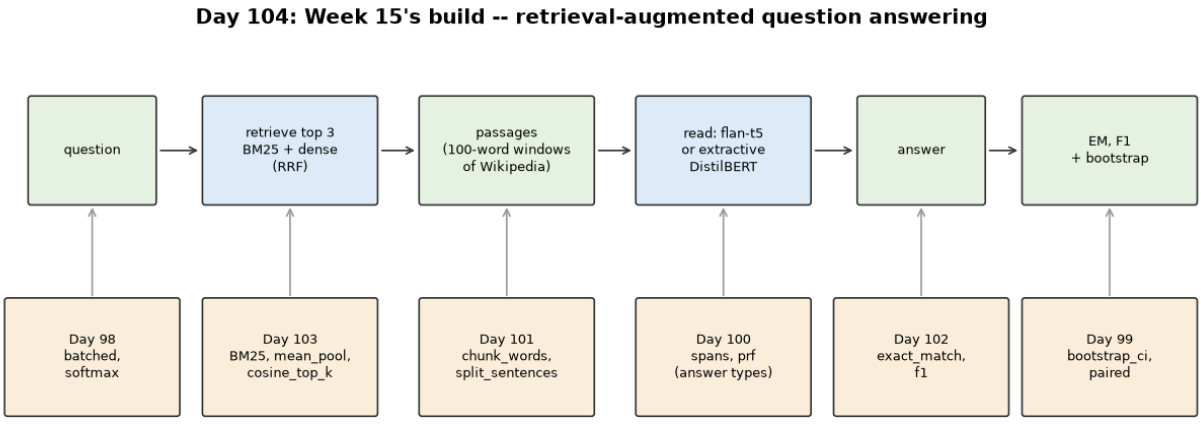
Retrieval-augmented generation (RAG) answers a question in two steps: **retrieve** the passages of a document collection most relevant to it (Day 103's search), then have a pretrained model answer **from those passages**. The model's knowledge comes from the documents at question time, not only from its training.

flan-t5-small is T5 further trained to follow instructions ('Answer the question using the passages'). It writes its answer, so unlike Day 102's extractive reader it is not limited to copying a span.

REAL-WORLD SCENARIO

An open-book exam: first find the right pages, then answer from them. A student with the book but no index, or an index but no reading skill, both do badly.

SYSTEM DIAGRAM



The pipeline (top) and the exercise part behind each step (bottom).

The build

Like Days 83, 90 and 97, every function from the week's exercise scripts is copied verbatim under a banner naming its source; Day 102's span search and question sample are imported. New today: passages, the two readers, the prompt, and the measurements.

How to read each section

Step	What it gives you
DEFINITION	What the idea is, in plain words. Start here.
REAL-WORLD SCENARIO	An everyday situation where the same idea shows up.
SPECIFICATION	The exact setup of the experiment.
IMPLEMENTATION	The real code from the script.
ACTUAL OUTPUT	What the script really printed, copied unchanged.
What it means	How to read the numbers, and what they do and do not show.
Why choose this? / Why NOT this?	When the idea is the right tool, and what it costs.

1. The build: passages and the index

DEFINITION — start here, no prior knowledge assumed

Paragraphs are cut into **passages**, the unit retrieved and handed to a reader: windows of 100 words overlapping by 25 (`chunk_words`), or whole sentences packed up to 100 words (`split_sentences`).

REAL-WORLD SCENARIO

Cutting a long book into index cards small enough to hand to a reader.

SPECIFICATION

Documents	every distinct SQuAD v1.1 validation paragraph (48 Wikipedia articles).
Questions	the first 500 of Day 102's 1,000 random validation questions.
Models	MiniLM retriever (Day 103), DistilBERT-SQuAD reader (Day 102), flan-t5-small generator.

IMPLEMENTATION

```
def word_passages(docs):
    out = [(p, i) for i, d in enumerate(docs) for p in chunk_words(d, CHUNK, OVERLAP)]
    return [p for p, _ in out], np.array([i for _, i in out])
```

ACTUAL OUTPUT

```
Every function in section 1 is copied verbatim from a Week 15 exercise script (a banner names each one); Day
    102's span search and question sample are imported.
documents: 2,067 SQuAD validation paragraphs (123 words on average, 4.9 sentences)
word passages: 3,681 windows of 100 words overlapping by 25 (chunk_words); sentence passages: 3,627, whole
    sentences up to 100 words (split_sentences)
questions: 500; a gold answer appears in at least one word passage for 1.000 of them (an answer can be cut by
    a window edge, but the overlap usually saves it)
Actual output from this lesson's run.
```

What it means

The 2,067 paragraphs become 3,681 word passages or 3,627 sentence passages. Every one of the 500 questions has a gold answer inside some word passage, so cutting lost none of them.

WHY CHOOSE THIS?

Short passages fit a small reader's input and keep the answer near the question's words.

WHY NOT THIS? (TRADEOFFS)

A window edge can cut an answer in two; the overlap usually, not always, saves it.

2. Retrieval: is the answer in the top k passages?

DEFINITION — start here, no prior knowledge assumed

BM25, dense and the RRF hybrid over both kinds of passage. A passage counts as a hit if it contains a gold answer string. That test is lenient: a short answer such as a year can appear in passages about something else.

REAL-WORLD SCENARIO

Checking whether the answer is somewhere on the cards you handed over, before blaming the reader.

IMPLEMENTATION

```
    S_b = np.array([bm.scores(q) for q in qs])
    idx_d, _ = cosine_top_k(Qv, Pv, 10)
    for mname, idx in (("BM25", np.argsort(-S_b, axis=1)[:, :10]), ("dense", idx_d),
                       ("hybrid", np.argsort(-rrf(S_b, Qv @ Pv.T), axis=1)[:, :10])):
        hit = np.array([[contains(P[j], g) for j in row] for row, g in zip(idx, golds)])
        res[(pname, mname)] = [hit[:, :k].any(1).mean() for k in (1, 3, 5, 10)]; top[(pname,
            mname)] = idx
```

ACTUAL OUTPUT

```
answer-containing passage within the top k (indexing and searching both passage types: 130 s)
passages  retriever     k=1     k=3     k=5    k=10
word      BM25        0.752   0.884   0.918   0.938
word      dense       0.634   0.846   0.880   0.928
word      hybrid      0.754   0.888   0.934   0.966
sentence  BM25        0.746   0.872   0.918   0.934
sentence  dense       0.682   0.824   0.882   0.928
sentence  hybrid      0.768   0.898   0.932   0.970
word passages, hybrid, top 3 as a set (day100_exercise.py's prf): precision 0.365, recall 0.049
    (of 11,111 answer-holding passages; several can hold one answer)
Actual output from this lesson's run.
```

What it means

The hybrid over word passages puts an answer in the top 3 for 0.888 of questions and in the top 10 for 0.966; the two passage types are close. The set view (Day 100's `prf`) shows the leniency: 11,111 passages hold some question's answer string, so recall over all of them is only 0.049 although one good passage is enough.

WHY CHOOSE THIS?

Retrieval recall is the ceiling on everything after it.

WHY NOT THIS? (TRADEOFFS)

String matching over-counts short answers; true relevance would need labels.

3. End to end: closed book, RAG, oracle

DEFINITION — start here, no prior knowledge assumed

Five systems on the same 500 questions: flan-t5 with the question alone (**closed book**); flan-t5 and the extractive reader on the top 3 hybrid passages (**RAG**); and both readers handed the right paragraph (**oracle**, an upper bound). A prompt longer than T5's 512-token input has its passages trimmed, never the question.

REAL-WORLD SCENARIO

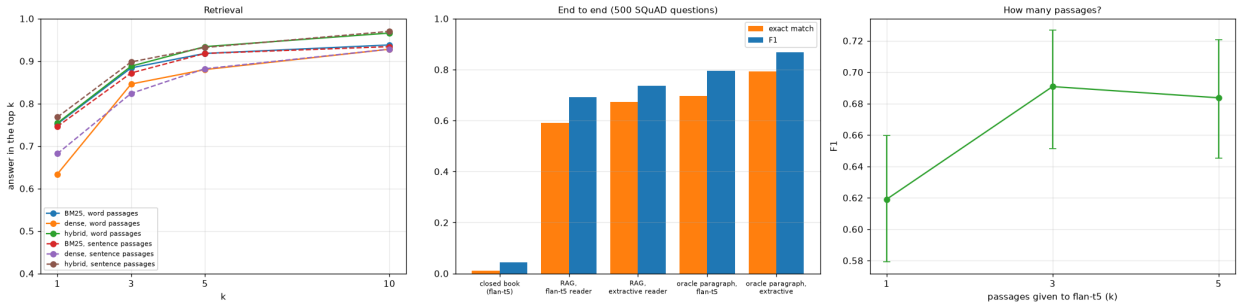
The same exam taken from memory, with the cards an index found, and with the exact page.

IMPLEMENTATION

```
def fit_prompt(gtok, q, passages, limit=512):
    rest = len(gtok(PROMPT.format(ctx="", q=q))["input_ids"])
    ids = gtok("\n\n".join(passages), add_special_tokens=False)["input_ids"]
    cut = len(ids) > limit - rest
    ctx = gtok.decode(ids[:limit - rest]) if cut else "\n\n".join(passages)
    return PROMPT.format(ctx=ctx, q=q), cut
```

ACTUAL OUTPUT

```
google/flan-t5-small: 76,961,152 weights; prompt: 'Answer the question using the passages.\n\n{ctx}\n\nQuestion:
    {q}\nAnswer:'
system                          exact match      F1     F1 95% interval  s per question
closed book (flan-t5)                 0.010   0.043      [0.031, 0.057]            0.03
RAG, flan-t5 reader                   0.590   0.691      [0.651, 0.727]            0.16
RAG, extractive reader                0.672   0.736      [0.702, 0.773]            0.15
oracle paragraph, flan-t5             0.696   0.794      [0.762, 0.826]            0.12
oracle paragraph, extractive          0.792   0.868      [0.842, 0.894]            0.07
paired differences in F1 (day99_exercise.py): mean, 95% interval, share of resamples ahead
  RAG, flan-t5 reader           minus closed book (flan-t5)  +0.648 [+0.609, +0.685] ahead in 1.000
  RAG, extractive reader        minus RAG, flan-t5 reader    +0.046 [+0.013, +0.081] ahead in 0.996
  oracle paragraph, flan-t5     minus RAG, flan-t5 reader    +0.103 [+0.070, +0.137] ahead in 1.000
  oracle paragraph, extractive  minus RAG, extractive reader +0.131 [+0.099, +0.162] ahead in 1.000
three questions, every system:
  Q: What did Low suggest using Apollo 8 for instead of simply orbiting the Earth's surface?   gold: 'orbit the Moon'
     closed book (flan-t5)         'a heliocentric orbit'
     RAG, flan-t5 reader           'the D mission'
     RAG, extractive reader        'eliminating the E mission'
     oracle paragraph, flan-t5     'deferring the D mission to the next mission in March 1969, a'
     oracle paragraph, extractive  'eliminating the E mission'
  Q: How did Luther's writings sound as he became less healthy?   gold: 'harsher'
     closed book (flan-t5)         'a snort'
     RAG, flan-t5 reader           'at least partly the product of a declining state of mind'
     RAG, extractive reader        'partly the product of a declining state of mind'
     oracle paragraph, flan-t5     'short-tempered'
     oracle paragraph, extractive  'harsher'
  Q: When might starch grains become overly large?   gold: 'Under conditions such as high atmospheric CO2 concentrations'
     closed book (flan-t5)         'a year'
     RAG, flan-t5 reader           'Under conditions such as high atmospheric CO2 concentrations'
     RAG, extractive reader        'Under conditions such as high atmospheric CO2 concentrations'
     oracle paragraph, flan-t5     'Under conditions such as high atmospheric CO2 concentrations'
     oracle paragraph, extractive  'Under conditions such as high atmospheric CO2 concentrations'
Actual output from this lesson's run.
```



Left: retrieval recall at k. Middle: exact match and F1 of the five systems. Right: flan-t5's F1 against the number of passages, with 95% intervals.

What it means

Retrieval is what makes the small model useful: closed book it gets 0.010 exact match; with passages 0.590 (F1 +0.648 higher). The extractive reader is a little better than the generator here (+0.046 F1), likely because SQuAD's answers are spans by construction. The gap to the oracle (+0.103 and +0.131) is the price of retrieving instead of being handed the paragraph. The second example shows that price: handed the right paragraph the extractive reader finds 'harsher'; given the retrieved passages it does not.

WHY CHOOSE THIS?

Small models become capable on a document collection they never trained on.

WHY NOT THIS? (TRADEOFFS)

Every answer depends on retrieval, and the costs add up per question.

4. Where RAG fails: retrieval or reading?

DEFINITION — start here, no prior knowledge assumed

For every question answered with F1 = 0, whether any of the top 3 passages contained a gold answer: if not, retrieval failed; if so, the reader did. And the extractive reader's confidence: a softmax over the 3 passages' best span scores (Day 98's part).

REAL-WORLD SCENARIO

Deciding whether the student lacked the right page or misread it.

ACTUAL OUTPUT

```
the top 3 passages hold a gold answer for 0.888 of questions
  RAG, flan-t5 reader    : F1 = 0 on 111 questions -> passages lacked the answer 46, reader missed it although
      present 65; F1 when the answer was retrieved 0.771, when it was not 0.056
  RAG, extractive reader : F1 = 0 on 102 questions -> passages lacked the answer 49, reader missed it although
      present 53; F1 when the answer was retrieved 0.823, when it was not 0.047
  extractive reader's passage probability (softmax over 3 passages, so never below 1/3):
    [0.33, 0.70): 33 questions, exact match 0.545
    [0.70, 0.95): 81 questions, exact match 0.457
    [0.95, 1.00): 386 questions, exact match 0.728
Actual output from this lesson's run.
```

What it means

Of flan-t5's 111 complete misses, 46 had no answer in the passages and 65 had one and still missed; the extractive reader's 102 split 49 and 53. Both halves need work. When the answer was retrieved, F1 is 0.771 and 0.823; when it was not, it is near zero. The reader's passage probability helps only at the top: exact match is 0.728 in the highest band, but the two lower bands are not in order (0.545 on 33 questions, 0.457 on 81 questions), so below 0.95 it says little.

WHY CHOOSE THIS?

Points improvement effort at the right component.

WHY NOT THIS? (TRADEOFFS)

The 'contains the answer' test is lenient, so some 'reader' failures may be retrieval failures.

5. How many passages?

DEFINITION — start here, no prior knowledge assumed

The generator given the top 1, 3 or 5 passages. More passages raise the chance the answer is there, but also add distraction and may not fit: then the passages are trimmed.

REAL-WORLD SCENARIO

Handing a reader one card, three, or five: more is not always better.

ACTUAL OUTPUT

```
  k = 1: exact match 0.534, F1 0.619 [0.579, 0.660]; passages shortened to fit 512 tokens: 0.000
  k = 3: exact match 0.590, F1 0.691 [0.651, 0.727]; passages shortened to fit 512 tokens: 0.016
  k = 5: exact match 0.578, F1 0.684 [0.645, 0.721]; passages shortened to fit 512 tokens: 0.856
Actual output from this lesson's run.
```

What it means

Three passages beat one (0.691 against 0.619 F1). Five do not help (0.684): 0.856 of those prompts had to be trimmed to 512 tokens, so the extra passages were partly cut away, and a small model may also be distracted by more text (not separated here).

WHY CHOOSE THIS?

A clear choice for this model and passage size.

WHY NOT THIS? (TRADEOFFS)

The best k depends on the reader's input length; a bigger model would change it.

6. Which answers are hardest?

DEFINITION — start here, no prior knowledge assumed

Each question's first gold answer tagged by Day 100's NER model; the longest entity in it gives the answer's type (Day 100's `spans`). Numbers, dates and phrases are 'not an entity'.

REAL-WORLD SCENARIO

Some questions want a name, others a description: they need different skills.

ACTUAL OUTPUT

```
answer type     questions         closed book        RAG, flan-t5     RAG, extractive     oracle, flan-t5  oracle, extractive
not an entity         317                0.035               0.672               0.736               0.779               0.873
ORG                    57                0.061               0.852               0.789               0.873               0.853
MISC                   44                0.011               0.636               0.692               0.774               0.819
PER                    43                0.031               0.752               0.811               0.827               0.922
LOC                    39                0.127               0.603               0.629               0.792               0.850
(mean F1 per type; numbers, dates and phrases are 'not an entity')
Actual output from this lesson's run.
```

What it means

Most answers (317 of 500) are not named entities. RAG improves every type over the closed book. With about 40-57 questions per entity type the differences between types are rough; the generator's best type is organisations (0.852). Two earlier runs gave slightly different counts here, because ties between equally long entities depended on Python's set order; the script now sorts them, and two runs with different hash seeds agree exactly.

WHY CHOOSE THIS?

Connects the week's NER day to the QA system's errors.

WHY NOT THIS? (TRADEOFFS)

Small groups, and NER on short answer strings is imperfect.

Interview preparation — technical questions

Q: What are the parts of a RAG system?

A: Chunking, an index, a retriever, a reader or generator, and an evaluation.

Q: How do you diagnose a RAG failure?

A: Check whether the retrieved passages contain the answer: retrieval vs reading.

Q: What happens when the prompt is too long?

A: Trim the passages, never the question; here 0.856 of k = 5 prompts needed it.

Q: Extractive or generative reader?

A: Extractive was better here (+0.046 F1); generative can combine and rephrase.

Q: What does the oracle measure?

A: The reader's ceiling with perfect retrieval: +0.103 to +0.131 F1 here.

Interview preparation — theoretical questions

Q: Why does retrieval help a small model so much?

A: The answer is in the passage; the model only has to find and phrase it.

Q: Why not give the model every passage?

A: Input limits, cost, and distraction: k = 5 did not beat k = 3.

Q: Why is 'passage contains the answer' a weak relevance test?

A: Short answers appear in unrelated passages.

Q: How would you make RAG say 'I don't know'?

A: Threshold the reader's score; Day 102 showed it separates right and wrong paragraphs.

Key takeaway and what's next

KEY TAKEAWAY

Retrieval-augmented QA is the week's parts in a line: cut, index, retrieve, read, score. It turns a small model that knows almost nothing about these articles (F1 0.043) into one that answers most questions (0.691 generative, 0.736 extractive). Its failures split between retrieval and reading, so both have to be measured. Week 15 is complete.

- Retrieval (hybrid, word passages): answer in top 3 for 0.888, top 10 for 0.966.
- F1: closed book 0.043; RAG 0.691 (flan-t5) and 0.736 (extractive); oracle 0.794 and 0.868.
- Complete misses: flan-t5 46 retrieval + 65 reading; extractive 49 + 53.
- k: 1 -> 0.619, 3 -> 0.691, 5 -> 0.684 F1.

Suggested watch order, if starting from zero

- 1. What Is RAG? Retrieval-Augmented Generation Explained Simply (Redis) — what RAG is and why it exists.
- 2. RAG Explained in 5 Minutes (KnowHive) — the pipeline in five minutes.

Day 105 preview

- **Week 16: classical ML, starting with decision trees from scratch.** Entropy and Gini on a real dataset, and why one tree is unstable.